Evaluation of Cognitive Appraisal Signals in Large Language Models: A Bilingual Study

1st Jonathan Fernández-Córdova
Departamento de Sistemas y Computación, TecNM
Instituto Tecnológico de Veracruz
Veracruz, México
l205020480@veracruz.tecnm.mx
ORCID: 0009-0007-5609-2556

2nd Rafael Rivera-López
Departamento de Sistemas y Computación, TecNM
Instituto Tecnológico de Veracruz
Veracruz, México
rafael.rl@veracruz.tecnm.mx
ORCID: 0000-0002-5254-4195

3rd Martín Hernández-Ordoñez
Departamento de Metal-Mecánica, TecNM
Instituto Tecnológico de Veracruz
Veracruz, México
martin.ho@veracruz.tecnm.mx
ORCID: 0000-0002-3740-5736

4th Paulina Galindo-Garrido
Departamento de Estudios de Posgrado e Investigación, TecNM
Instituto Tecnológico de Apizaco
Apizaco, Tlaxcala, México
ggpaulina31@gmail.com
ORCID: 0009-0004-4474-6952

***Abstract*—Large language models (LLMs) may exhibit variations in their behavior in response to modifications in instruction phrasing. This work evaluates whether linguistic signals associated with cognitive appraisal can influence the performance of a language model and whether such behavior remains consistent across different languages. A controlled experiment has been designed using Qwen 14B and 385 questions evaluated in English and Spanish. Four conditions were established: *Original*, *Anxiety*, *Importance*, and *Consequences*, using ten linguistic variants for each experimental signal. The same questions and sampling seed were used in both languages. Performance was evaluated using *accuracy*, 95% confidence intervals based on the Wilson method, McNemar's tests, and logistic regression to analyze the effects of language, condition, and their interaction. The results show that the evaluated cognitive appraisal signals do not produce statistically significant differences in model performance. On the other hand, there exists a difference between English and Spanish ($p < 0.001$), with English consistently outperforming Spanish across all tested conditions. Moreover, the interaction between language and condition did not reach statistical significance ($p = 0.6838$). These results indicate that, under the tested scenarios, Qwen 14B does not appear particularly sensitive to the evaluative phrasing used. However, language itself does seem to play a clear and consistent role in its performance.**

***Index Terms*—Large Language Models, LLMs, Appraisal Theory, emotional prompting, multilingual evaluation, Qwen 14B, MMLU.**

I. Introduction

Large language models (LLMs) have demonstrated advanced capabilities for solving tasks involving knowledge, comprehension, and reasoning through instructions expressed in natural language. The behavior of these models does not depend solely on the content of a task, but may also be affected by how instructions are formulated. This observation has motivated the study of different *prompt engineering* strategies, including methods that incorporate additional information into the main task content [2], [3].

Within this line of research, *emotional prompting* has investigated the incorporation of expressions associated with emotional states into instructions directed at language models. In particular, EmotionPrompt reported that certain emotional signals can modify LLM behavior and lead to performance improvements on specific tasks [1]. These findings raise the possibility that contextual information that is not directly part of a problem may influence the response generated by a model.

A related theoretical perspective is *Appraisal Theory*, which proposes that emotional responses are related to the cognitive evaluation that an individual performs regarding a situation. Within this framework, the appraisal of aspects such as the relevance or importance of an event and its potential consequences may contribute to the generation of emotional responses [4], [7]. This perspective makes it possible to study emotional expressions not only as superficial characteristics of language, but also as signals that convey information about the appraisal of a situation.

From this perspective, a relevant question arises for language models: whether linguistic signals associated with cognitive appraisal processes produce systematic changes in LLM performance. This question is particularly relevant in multilingual scenarios, where the same model may exhibit differences in performance depending on the language used to express a task. Recent work on multilingual evaluation has highlighted the importance of conducting controlled comparisons across languages to investigate the capabilities of language models [5], [6].

In this work, we experimentally evaluate whether three types of signals associated with cognitive appraisal, referred to as *Anxiety*, *Importance*, and *Consequences*, affect the per-

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formance of a language model. The study uses Qwen 14B and a set of 385 questions selected in a reproducible manner. The same questions are evaluated in English and Spanish while maintaining the same sampling seed and an equivalent experimental structure. For each condition, ten linguistic variants of the stimulus are used to reduce dependence on a single textual formulation. The *Original* condition, with no additional stimulus, is used as the baseline.

The main goal is to determine whether cognitive appraisal signals produce statistically significant differences in model performance and to establish whether such effects behave differently between English and Spanish. To this end, performance is quantified using *accuracy*, with 95% confidence intervals calculated using the Wilson method. Additionally, statistical tests are employed to compare the experimental conditions and assess the potential interaction between language and cognitive appraisal signals.

The results show that the *Anxiety*, *Importance*, and *Consequences* conditions do not produce a statistically significant change in Qwen 14B performance under the experimental conditions studied. In contrast, a significant performance difference is observed between English and Spanish. Furthermore, the interaction analysis provides no evidence that language significantly modifies the effect of the evaluated signals. These results make it possible to distinguish between the effect of cognitive appraisal signals and performance differences associated with language.

II. Related Work

The behavior of language models can be affected by modifications in the formulation of instructions, even when the main content of the task remains unchanged. This property has motivated the development of different *prompt engineering* strategies, including techniques aimed at inducing specific reasoning or behavioral patterns through additional contextual information [2].

In particular, EmotionPrompt investigated the incorporation of emotional stimuli into language model instructions and reported performance improvements across a set of evaluation tasks [1]. This work provides evidence that prompt characteristics that are not directly part of the task content can influence model responses. However, these findings do not necessarily establish that all types of emotional or cognitive signals produce consistent effects across different models and experimental conditions.

From a psychological perspective, *Appraisal Theory* establishes that emotional responses are related to the evaluation an individual performs regarding certain aspects of a situation [4], [7]. These aspects include the relevance of an event and the appraisal of its potential consequences. In the present work, these concepts are used as a framework for constructing linguistic stimuli associated with *Anxiety*, *Importance*, and *Consequences*, without assuming that the model possesses internal emotional states equivalent to those of humans.

Furthermore, multilingual evaluation of language models has shown that performance may vary across languages and that comparisons require controlled experimental conditions. Recent benchmarks such as MMLU-ProX emphasize the need to use comparable tasks across languages to analyze the multilingual capabilities of LLMs [5], while broader multilingual evaluations have documented performance differences across languages [6].

Unlike these works, the present study combines the evaluation of cognitive appraisal signals with a controlled bilingual comparison. The same 385 questions are used in English and Spanish, while maintaining the same question selection and sampling seed. This enables the effects of appraisal conditions to be analyzed separately and determines whether such effects exhibit different behavior across the two languages.

III. Methodology

A. Experimental Design

A quantitative, comparative, and black-box approach was adopted to evaluate the effect of linguistic signals associated with cognitive appraisal on the performance of a language model. The experimental design considered two factors: language and condition. The language factor had two levels, English and Spanish, while the condition factor had four levels: *Original*, *Anxiety*, *Importance*, and *Consequences*.

For each combination of language and condition, 385 questions were used, resulting in 1,540 evaluations per language and 3,080 evaluations across the complete experiment. The questions were selected using a fixed random seed so that the same questions were evaluated in both English and Spanish. This configuration enables a controlled comparison in which the task content remains constant, while language and stimulus condition constitute the factors of interest.

B. Model and Evaluation Dataset

The experiments were conducted using Qwen-14B, a 14-billion-parameter transformer-based language model developed by Alibaba Cloud and executed locally under controlled generation settings [8]. Performance was evaluated using the MMLU (Massive Multitask Language Understanding) benchmark, which covers 57 tasks across diverse domains, including mathematics, history, and science. The corresponding Spanish version was evaluated using the same question selection and sampling seed to ensure direct comparability between languages.

Each question contains four answer choices, identified as A, B, C, and D. The model was instructed to provide its answer using a controlled format, allowing the selected option to be automatically identified. The generation temperature was fixed at zero to reduce stochastic variability across experimental conditions. A response was considered correct when the generated option matched the reference answer of the corresponding question.

C. Stimulus Construction

The experimental conditions were constructed around three appraisal-related dimensions: *Anxiety*, *Importance*, and *Consequences* [4], [7]. The *Original* condition served as the baseline

TABLE I
REPRESENTATIVE EXAMPLES OF EXPERIMENTAL STIMULI IN ENGLISH AND SPANISH

Condition	English	Spanish
Original	No additional stimulus; the original MMLU question is presented directly.	Sin estímulo adicional; se presenta directamente la pregunta original de MMLU.
Anxiety	I am very nervous about the outcome.	Estoy muy nervioso por el resultado.
Importance	This task is very important.	Esta tarea es muy importante.
Consequences	There will be significant consequences if the answer is incorrect.	Habrá consecuencias significativas si la respuesta es incorrecta.

and contained no additional linguistic stimulus; the original benchmark question was presented directly to the model.

The three dimensions were selected to represent distinct forms of task-related appraisal: affective concern about the outcome, perceived importance of the task, and anticipated consequences associated with an incorrect response. *Anxiety* was operationalized through expressions of nervousness, concern, uneasiness, tension, or apprehension regarding the outcome of the task. *Importance* was operationalized through statements emphasizing the relevance or importance of the task, question, or outcome. *Consequences* was operationalized through statements referring to potential consequences associated with providing an incorrect answer. These conditions were designed to provide distinct forms of task-related contextualization while leaving the underlying benchmark question unchanged.

Each non-baseline condition was represented by ten linguistic variants. The ten variants for each condition were formulated in both English and Spanish, with the objective of preserving their intended meaning and linguistic function across languages. Representative examples of the four experimental conditions are presented in Table I. The use of multiple linguistic variants reduces dependence on a single wording and provides a more robust operationalization of each experimental condition.

D. Representational Validation of the Stimuli

Before the performance analysis, a complementary validation of the stimuli was conducted using vector representations (*embeddings*). The embeddings corresponding to the ten variants of each of the three experimental conditions were projected using Principal Component Analysis (PCA).

The first two principal components jointly explained 62.3% of the observed variance, with 35.7% corresponding to the first component and 26.6% to the second. The projection showed distinct clusters for *Anxiety*, *Importance*, and *Consequences*, indicating a representational separation among the three stimulus categories.

This validation was used as complementary evidence that the categories exhibit representational differences and not as evidence of the existence of internal emotional states in the model. Consequently, PCA is interpreted solely as a verification of the differentiation among the stimuli used in the experiment.

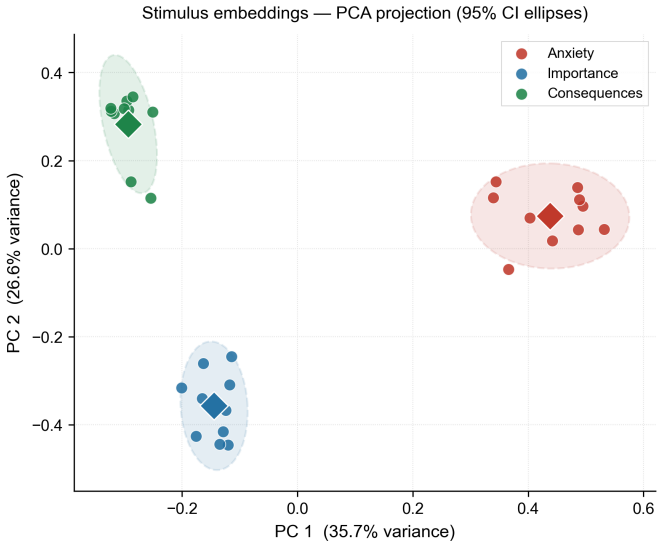


Fig. 1. PCA projection of the embeddings corresponding to the *Anxiety*, *Importance*, and *Consequences* variants. The first two components jointly explain 62.3% of the variance.

E. Evaluation Procedure

Each question was evaluated under the four experimental conditions in each of the two languages. For the *Original* condition, the question was presented without additional linguistic information. For the *Anxiety*, *Importance*, and *Consequences* conditions, one of the ten corresponding linguistic variants was incorporated into the prompt.

The overall procedure can be expressed as follows:

$$\text{Question}+\text{Stimulus} \rightarrow \text{Qwen 14B} \rightarrow \text{Response} \rightarrow \text{Evaluation} \tag{1}$$

The generated response was automatically processed to identify the selected answer choice. This choice was then compared with the reference answer from the benchmark, and a binary outcome was assigned: 1 for a correct response and 0 for an incorrect response.

F. Variables and Metrics

The primary independent variable was the experimental condition, while language was incorporated as a second factor. The dependent variable was accuracy, defined as the proportion of correct responses relative to the total number of questions evaluated.

$$\text{Accuracy} = \frac{\text{Correct responses}}{\text{Total questions}} \tag{2}$$

For each combination of language and condition, 95% confidence intervals were calculated using the Wilson method for binomial proportions. Additionally, the number of responses that could not be automatically interpreted was recorded in order to distinguish performance errors from output formatting issues.

TABLE II
QWEN 14B PERFORMANCE BY LANGUAGE AND CONDITION.

Language	Condition	Correct	Accuracy (%)
English	Original	209/385	54.29
English	Anxiety	212/385	55.06
English	Importance	202/385	52.47
English	Consequences	196/385	50.91
Spanish	Original	168/385	43.64
Spanish	Anxiety	162/385	42.08
Spanish	Importance	158/385	41.04
Spanish	Consequences	169/385	43.90

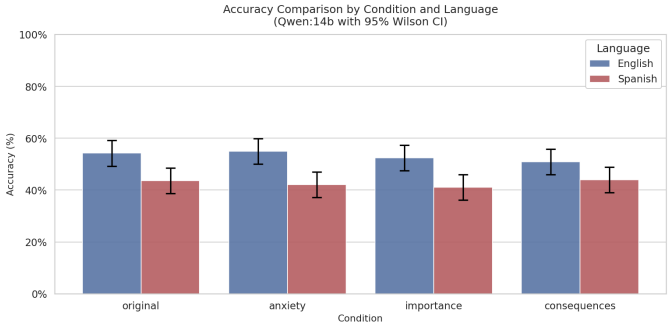


Fig. 2. Qwen 14B accuracy by condition and language with 95% Wilson confidence intervals.

G. Statistical Analysis

Because the same questions were used under the different experimental conditions, comparisons between the *Original* condition and each stimulus condition were performed while accounting for the paired nature of the observations. McNemar's test was used to determine whether statistically significant differences existed in the proportion of correct responses between conditions.

To jointly analyze the language and condition factors, a binary logistic regression model was fitted using response correctness as the dependent variable. The model included the main effects of language and condition, as well as their interaction terms. The interaction terms allowed us to assess whether the effect of the cognitive appraisal stimuli varied significantly between English and Spanish.

The statistical significance level was set at $\alpha = 0.05$. Thus, the analysis enables us to distinguish between performance differences associated with language and potential changes produced by cognitive appraisal signals.

IV. RESULTS AND DISCUSSION

A. Performance by Condition

Table II presents the performance obtained by Qwen 14B under the four experimental conditions in English and Spanish. Each condition contains 385 questions. In English, the *Original* condition achieved an accuracy of 54.29%, while *Anxiety* obtained 55.06%, *Importance* 52.47%, and *Consequences* 50.91%. In Spanish, the *Original* condition achieved 43.64%, while *Anxiety*, *Importance*, and *Consequences* obtained 42.08%, 41.04%, and 43.90%, respectively.

The descriptive differences relative to the *Original* condition were small. In English, *Anxiety* showed an increase of 0.78 percentage points, whereas *Importance* and *Consequences* showed decreases of 1.82 and 3.38 percentage points, respectively. In Spanish, the three conditions showed changes of -1.56, -2.60, and +0.26 percentage points, respectively. These descriptive changes do not, by themselves, constitute evidence of an effect of the signals and were therefore subjected to statistical analysis.

B. Confidence Intervals

For each proportion, 95% confidence intervals were calculated using the Wilson method. The intervals show a considerable degree of overlap between conditions within each language, which is consistent with the absence of substantial differences in performance.

Figure 2 shows a more pronounced descriptive difference between languages than between conditions. Average performance in English was higher than that observed in Spanish across all evaluated conditions.

C. Effect of Cognitive Appraisal Signals

The paired comparisons between the *Original* condition and the experimental conditions did not provide sufficient statistical evidence to conclude that *Anxiety*, *Importance*, or *Consequences* modified the model's performance. Therefore, under the experimental conditions used in this study, no statistically significant effect of cognitive appraisal signals on Qwen 14B accuracy was observed.

This result is relevant in light of previous findings on *emotional prompting*. While EmotionPrompt showed that certain emotional stimuli can influence LLM performance, the results obtained in this study do not support generalizing this effect to the cognitive appraisal signals evaluated here [1]. One possible interpretation is that the presence of a signal associated with an emotional state or with the appraisal of a task is not, by itself, sufficient to produce a consistent change in model performance.

D. Effect of Language

In contrast to the experimental conditions, language showed a clear difference in performance. The average accuracy observed in English was higher than that obtained in Spanish. Considering all conditions jointly, performance was approximately 52.68% in English compared with 42.66% in Spanish.

The logistic regression model identified a statistically significant effect of language ($p < 0.001$), with a lower probability of a correct response for the Spanish evaluations compared with the English evaluations. This result indicates that language constitutes a relevant factor for Qwen 14B performance under the experimental design used in this study.

This finding is consistent with the need to conduct controlled multilingual evaluations, as a model may exhibit performance differences across languages even when the evaluated tasks are conceptually equivalent [5], [6].

E. Interaction Between Language and Condition

Finally, we evaluated whether language modified the effect of the cognitive appraisal signals using a logistic regression model with interaction terms between language and condition. The likelihood ratio test yielded a $\chi^2 = 1.4934$ statistic with 3 degrees of freedom and a $p = 0.6838$ value.

Therefore, there is no statistically significant evidence of an interaction between language and condition. In other words, although overall performance was significantly different between English and Spanish, the data do not provide sufficient evidence to conclude that the effect of *Anxiety*, *Importance*, or *Consequences* depends on the language.

Overall, the results reveal a distinction between two phenomena: language is associated with significant differences in performance, whereas the evaluated cognitive appraisal signals show neither a statistically significant effect nor a significant interaction with language. Therefore, the results do not support the hypothesis that these signals produce a generalizable improvement in Qwen 14B performance under the conditions studied.

V. Conclusion

This study evaluated whether linguistic signals associated with cognitive appraisal modify the performance of Qwen 14B and whether their effect varies between English and Spanish. Using a controlled bilingual experimental design with 385 questions per condition, three appraisal-related signals were evaluated: *Anxiety*, *Importance*, and *Consequences*.

The results indicate that none of the evaluated cognitive appraisal signals produced a statistically significant modification of model accuracy relative to the *Original* condition. Although small descriptive differences were observed across conditions, the statistical analyses did not provide sufficient evidence to conclude that the evaluated signals systematically improve or decrease model performance.

In contrast, language was associated with a statistically significant difference in performance. Qwen 14B achieved higher accuracy in English than in Spanish across all experimental conditions, with average accuracies of approximately 52.68% and 42.66%, respectively. However, the interaction analysis did not provide evidence that language significantly modified the effect of the cognitive appraisal conditions ($\chi^2 = 1.4934$, $p = 0.6838$).

Therefore, the findings suggest that, under the experimental conditions studied, language represents a more relevant source of performance variation than the evaluated cognitive appraisal signals. The results also indicate that findings previously reported for emotional prompting should not be assumed to generalize directly to other types of evaluative or appraisal-based signals. Instead, their effect appears to depend on the specific stimulus, model, task, and evaluation setting.

VI. Future Work

Future work will extend the proposed evaluation in three directions. First, the experiment will be replicated across additional open-source language models with different architectures and parameter scales to determine whether the observed absence of a cognitive appraisal effect is specific to Qwen 14B or represents a broader behavior.

Second, the evaluation will be extended to additional multilingual benchmarks and languages to distinguish language-specific effects from more general multilingual performance differences. This would also allow the analysis to consider languages with different linguistic and cultural characteristics.

Finally, future experiments will investigate a broader range of appraisal dimensions, such as perceived controllability, certainty, novelty, goal relevance, and responsibility, as well as graded levels of stimulus intensity. Rather than treating each signal as a single categorical condition, graded appraisal levels could be introduced to determine whether the magnitude of the linguistic signal is associated with changes in model performance.

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